

INNOVATIVE MINING SYSTEMS

Unit-1

By

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Research and development definitions

- Creative and systematic work undertaken in order to increase the stock of knowledge – including knowledge of humankind, culture and society – and to devise new applications of available knowledge
- For an activity to be an R&D activity it must satisfy all five core criteria: 1 to be aimed at new findings (novel), 2 to be based on original, not obvious, concepts and hypotheses (creative), 3 to be uncertain about the final outcomes (uncertain), 4 to be planned and budgeted (systematic), and 5 to lead to results that could be possibly reproduced (transferable and/or reproducible)
- Basic research is experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular application or use in view.
 - Pure basic research is carried out for the advancement of knowledge, without seeking economic or social benefits or making an active effort to apply the results to practical problems or to transfer the results to sectors responsible for their application.
 - Oriented basic research is carried out with the expectation that it will produce a broad base of knowledge likely to form the basis of the solution to recognised or expected current or future problems or possibilities.

Research and development definitions

- Applied Research is original investigation undertaken in order to acquire new knowledge. It is, however, directed primarily towards a specific, practical aim or objective (including a client-driven purpose).

Activities under R & D

- Prototype development and testing
- Final construction and operation of a pilot plant where the primary objective is to make further improvements
- trial production where there is full scale testing and subsequent further design and engineering phases I to III of clinical trials.

Activities that do not meet the definition of R&D include:

- scientific and technical information services,
- general purpose or routine data collection,
- standardisation and routine testing feasibility studies (except into R&D projects) specialised,
- routine medical care literature reviews that are predominantly a summary of the current knowledge and findings of a particular R&D field or topic and do not include any critical assessment or report any new findings or original experimental work commercial,
- legal and administrative aspects of patenting, plant breeders rights, copyright, material transfer agreements or intellectual property licensing,
- option and assignment activities, and royalties routine computer programming, systems work or software maintenance stages of product development that do not meet the five R&D criteria above
- pre-production development
- market research construction of fully tested prototypes for marketing purposes after sales service and trouble-shooting industrial engineering and design for production purposes artistic performance or expression R&D financing and support services.

MAJOR ISSUES CONCERNING INDUSTRY

- Shorter life cycles of technology
- Intense national and international competition
- Global markets
- Quality, cost, delivery, after-sales service, international standards
- Intellectual property rights
- High risks in investment

MAJOR ISSUES CONCERNING INDUSTRY (Contd.)

- Shorter time between innovation and commercialization
- Research and development important
- Energy efficient technologies
- Environment friendly technologies
- Information and communication networks
- Socio-economic and socio-political factors
- Movement of capital across national boundaries

WHAT IS MEANT BY R&D

It is a set of activities carried out by an organization with the intention to develop new products or processes or services or improvements over the existing ones.

CHANGING PARADIGM IN R&D MANAGEMENT

- Market and buyers driven – organizations can get more out of their research by linking it more closely to market need and customers requirements rather than increased spending elsewhere
- R&D on 'business like' footing

CHANGING PARADIGM IN R&D MANAGEMENT (Contd.)

- Time and cost of R&D is important
- Expectations from R&D increased manifold
- R&D is getting increasingly more multi-disciplinary in nature
- Creativity and innovativeness of considerable significance

CHANGING PARADIGM IN R&D MANAGEMENT (Contd.)

- Managerial skills in R&D personnel increasingly required
- Borderless laboratories
- Intellectual property rights issues
- Changing from a strategy of 'hope' to a strategy of a more 'systematic, disciplined and accountable' R&D

CHANGING PARADIGM IN R&D MANAGEMENT (contd.)

- Impact of Information and Communication Technologies. These are impacting R&D management systems and practices in numerous ways, creating new threats and opportunities.
- Increased expectations of Head of the Institution

SPECIFIC ATTRIBUTES OF R&D PROJECTS

- Risky
- Difficult to assess time and cost of projects
- Major input – human resource
- Continuously evolving in nature
- Intellectual Property Rights – an important issue
- Financing of R&D

R& D MANAGEMENT

- At the Government Level
- At the Organizational Level

At Government Level

- Government as a
 - Promoter
 - Developer
 - Implementer
 - Facilitator

Instruments for Implementation

- Mandatory in Nature
 - Intellectual Property Rights Rules
 - Technology Import Policy
 - Pollution Control Laws
 - R&D Cess Act
 - Income Tax, Customs & excise duties
Legislations
- Guidelines/Support Measures ...
 - Financial Support to R&D Projects

Instruments for Implementation (Contd.)

- Incentives
- National Awards
- Technology Policy
- Five Year Plan Allocations
- Technopreneur Promotion Programme
- New Millennium Indian Technology Leadership Initiative (NIMITLI)
- Protecting innovators intellectual property
- Technology Parks and Incubators

Implementing Organizations at National level

- Ministry of S&T (DST,DSIR,DBT)
- Earth Sciences
- Ministry of Commerce & Industry
- MHRD
- Deptt. Of Space
- Deptt. Of Ocean Development
- Ministry of Defence
- Ministry of Railways
- Ministry of Agriculture

IMPORTANT ISSUES IN R&D MANAGEMENT

- Organizational planning for R&D activities (Short, medium and long term)
- R&D Project Management
- Human resource aspects
- Strategic alliances

IMPORTANT ISSUES IN R&D MANAGEMENT (Contd.)

- Pricing
- Joint Research
- Contract Research
- Marketing of R&D
- Measuring R&D Effectiveness

Science and Technology (S&T) System

The main components of the present S&T system in the country could be categorized under the following sectors-

- Central Government
- State Governments
- Higher Education Sector
- Public Sector Industry
- Private Sector Industry and
- Non-profit Institutions/scientific and industrial research organisations

SCIENTIFIC MINISTRIES - CENTRAL

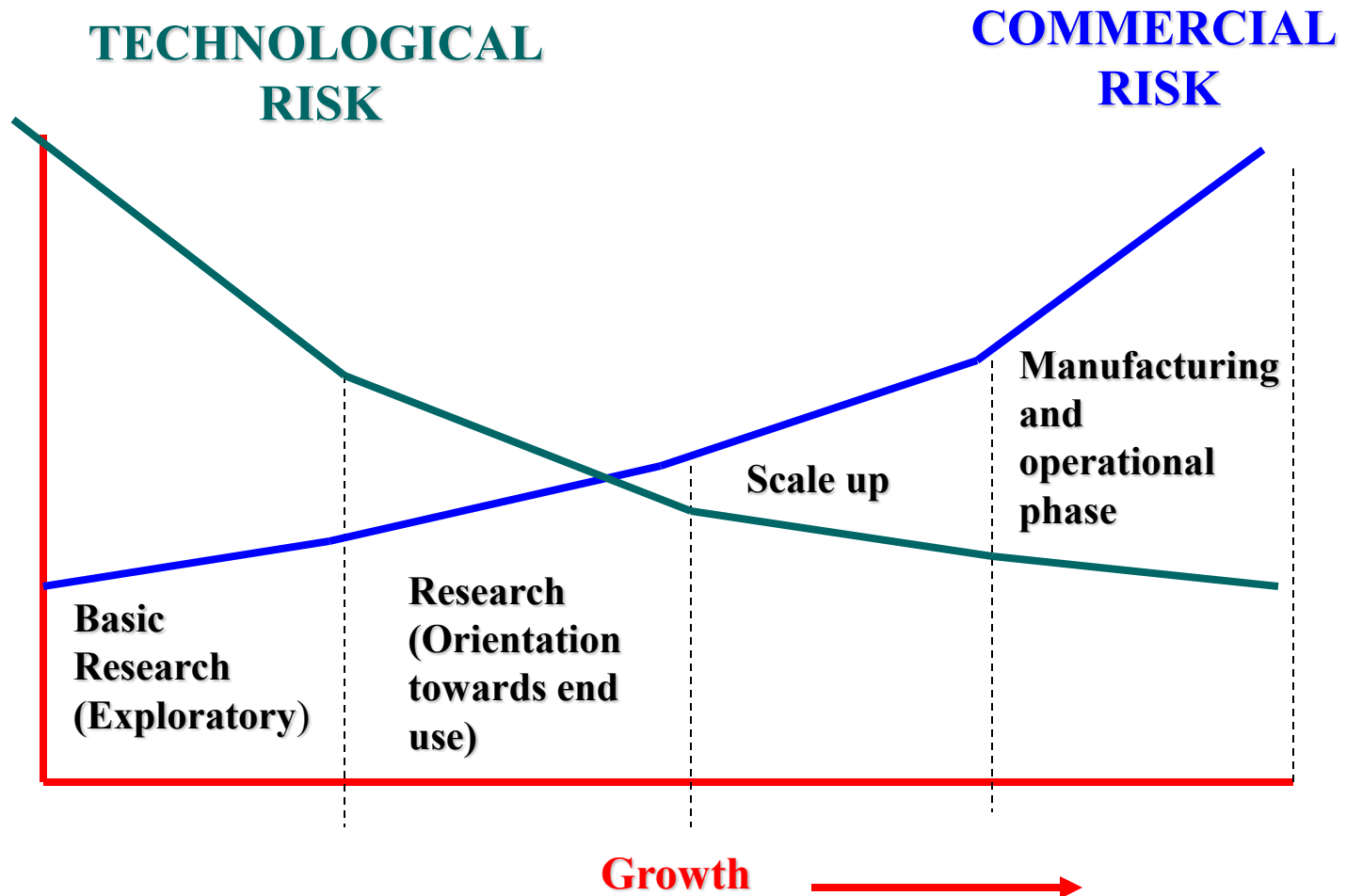
- ❖ **SCIENCE AND TECHNOLOGY**
- ❖ **SCIENTIFIC & INDUSTRIAL RESEARCH**
- ❖ **BIOTECHNOLOGY**
- ❖ **EARTH SCIENCES**
- ❖ **INFORMATION TECHNOLOGY**
- ❖ **NEW & RENEWABLE ENERGY**
- ❖ **ENVIRONMENT AND FORESTS**
- ❖ **ATOMIC ENERGY**
- ❖ **SPACE**
- ❖ **DEFENCE RESEARCH**
- ❖ **HEALTH RESEARCH**

PROFESSIONAL BODIES

(Government Funded)

- ❖ **INDIAN ACADEMY OF SCIENCES**
- ❖ **NATIONAL ACADEMY OF SCIENCES**
- ❖ **INDIAN NATIONAL SCIENCE ACADEMY**
- ❖ **INDIAN NATIONAL ACADEMY OF ENGINEERING**
- ❖ **INDIAN SCIENCE CONGRESS ASSOCIATION**

TECHNOLOGY DEVELOPMENT CHAIN



1. Basic Research (Exploratory Phase)

This is the starting point of the chain, driven primarily by scientific curiosity and the need to understand fundamental principles. Activities in this phase are exploratory in nature, often conducted in laboratories or academic settings.

- Objective:** Generate new knowledge and concepts.

- Risk profile:** *Very high technological risk* due to uncertainty in outcomes, but *low commercial risk* because market considerations are minimal.

- Output:** Scientific insights, theories, and early proof-of-concept ideas.

2. Research Orientation Towards End Use

In this phase, the focus shifts from pure science to solving specific practical problems. Research is oriented toward potential applications and end-user requirements.

- Objective:** Translate basic research outcomes into workable solutions.

- Risk profile:** Technological risk begins to reduce as feasibility improves, while commercial awareness starts to grow.

- Output:** Laboratory-scale prototypes, validated concepts, and performance benchmarks.

3. Development and Scale-Up

This is a critical transition stage where laboratory or pilot-scale results are expanded to industrially relevant scales. Engineering design, process optimization, and reliability testing dominate this phase.

- Objective:** Demonstrate that the technology can work consistently at larger scales.
- Risk profile:** Technological risk continues to decline, but **commercial risk increases** due to higher capital investment, cost uncertainties, and market competition.
- Output:** Pilot plants, demonstration units, and cost-performance data.

4. Manufacturing and Operational Phase

The technology reaches maturity and is deployed for routine production or field operation. Emphasis shifts to efficiency, quality control, regulatory compliance, and market acceptance.

- Objective:** Achieve stable, economical, and scalable operation.
- Risk profile:** Technological risk is relatively low, but **commercial risk is highest**, driven by market demand, pricing, competition, and long-term sustainability.
- Output:** Commercial products, operational systems, and revenue generation.

CASE STUDY-1

Non-Destructive, Transparent,
Quick and Accurate Gold
Analysis

- * Institute decides
- * Govt funds R&D
- * Developed and demonstrated successfully

NOT COMMERCIALISED



CASE STUDY-2

Ceramic Membrane Iron Removal Plant

(5000 ltrs/day)

- Institute decides
- Govt funds R&D
- Developed and demonstrated at large scale

**NO INDUSTRY PARTICIPATION BUT STILL
COMMERCIALISED**

The gives illustrative examples of demo projects as well as Replication projects taken up under the programme

Water purification plants for iron removal based on CGCRI Technology were set up in North Eastern States. A total of 16 plants are proposed to be set up out of which 8 have already been commissioned. Based on the success of the technology, Hon'ble Minister has announced setting up of 48 such plants to cover entire state of Arunachal Pradesh and Nagaland.

(Capacity : 5000 ltrs /day)



CASE STUDY – 3: HEPATITIS-B VACCINE

- INSTITUTE TAKES UP R&D
- INDUSTRY PITCHES IN
- GOVT PROVIDES FINANCIAL SUPPORT
- ALL WORK TOGETHER

COMMERCIALISED SUCCESSFULLY

Models for Valorizaion of IP

- Creating a Market Share in an established market
 - Licensing and R&D Collaboration
 - Breaking the monopoly of Multinationals
 - Solving Basic Problems of the Poor
 - Strategic and Public health
 - Attracting Contract research
 - Creating new opportunities through Image and Confidence Building
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Major IP Licensing Deal

- IMTECH's Clot specific Streptokinase Technology Licensed to NOSTRUM Pharmaceuticals nc., USA in July, 2006
 - Expected Returns: About 150 million US dollars
-subject to milestone payments
 - Royalties
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Success Stories in Licensing

- NCL-GE Alliance - originated in 1993.
 - GE supported the R&D at NCL
 - Alliance operating for over 9 years successfully and emerged as a paradigm in “relationship” management in R&D.
 - Cash flow to NCL from GE of around USD 8.5 m from assignment of 26 US patents over the period 1994-1995 to 2003-2004
 - Exposure to and training of NCL scientists to world class R&D management practices.
 - Building up of world class facilities and resources in NCL
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Success Stories in Licensing

- Attracting and hiring of talented young scientists
 - New contract research opportunities with multinational companies
 - Diffusion of ideas and generic methods developed to Indian industries
 - CSIR owns several patents based on generic ideas developed while interacting with GE having relevance to systems other than polycarbonates
 - **Led to Setting up of GE's R&D Centre at Bangalore**
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Success Stories in Licensing

United States Patent 6,893,479 (CSMCRI)

Integrated method for production of carrageenan and liquid fertilizer from fresh seaweeds

“An integrated method is developed to utilize to a maximum extent the fresh biomass of seaweeds such as *Kappaphycus alvarezii* that can be crushed to release sap and where the sap is useful as a potent **liquid fertilizer** after suitable treatment with additives and dilution while the residue is a superior raw material for extraction of κ-carrageenan, thereby enhancing the value of the seaweed. Other advantages of the invention include a reduced drying time and drying area to obtain the raw material for κ-carrageenan production in dry and storable form, a reduced cost of transporting and storing this raw material because of its lesser bulk, easier handling due to its free flowing granular nature, and its direct use for gel preparation in certain applications”

Success Stories in Licensing

- CSIR licenses Sea Weed Technology to Pepsico
 - A plant growth hormone and regulator is derived from fresh Sea Weed grown along the shoreline.
 - Liquid Sea Weed Fertilizer increases yields from same seed by 10% for wheat, 30% for peddy, 20-30% for fruits and vegetables to 35% for corn and chana, BT cotton 20%, Banana for 19%.
 - Patent application filed in India and abroad
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- Licensing of IICT's anti-cancer patent portfolio to a US Biopharmaceutical Co
- Returns: Rs. 2.5 Crores (subject to milestones) +royalties

Current & Future IP Strategy

- To plan and design strategic patent portfolios for short, medium and long term development requirement of national economy
 - To partner with national and international players to develop strategic IP portfolios for national economic needs
 - To focus on commercially and strategically important inventions
 - Valorization of existing IP Portfolios
 - Formation of a separate entity for licensing / spin-offs etc.
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Challenges

- How to implement the Patent strategy to avoid undesired duplication of research efforts
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Analyzing Patent Information

- Assessing State of the Art
 - Conducting Patentability Search
 - Patent mapping for ascertaining Value of IP
 - Studying existing claims
 - Freedom to Operate Analyses
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Analyzing Patent Information

- Anticipating other fields of use
 - Cross-discipline review
 - Design-around analysis
 - Evaluating competitors' IP sPace
 - Looking at possible infringement issues
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Analyzing Patent Information

Citation analysis: Finding key patents

- Identifying patents frequently referenced in other patents
 - Shows who's actively patenting in an area
 - Citation tree
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Integrating Patent Search Results with Other Market Research Data

Identifying companies developing similar products

- Internet
 - Trade Fairs
 - Industry associations
 - Subject-matter experts
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